Velcro®, a fastener with tiny hooks that catch loops, was inspired by the hooks of natural burrs catching on a dog's coat.

Mimicking nature

Nature is a great place to find ideas. Scientists try to copy features that help living things solve problems, such as the sticky pads that let lizards climb walls, and the way plants capture the sun's energy.

Velcro® is now used for many different things, from fastening shoes to holding objects in place inside the International Space Station!

Flexible laminate packaging is made from layers of plastic, paper, and metal.

Different types of synthetic rubber, which is used to make things such as tires, are made from unique combinations of materials.

The best of both

Composite materials combine two or more materials with different properties. For example, a light material and a strong material could make a strong, lightweight composite.

Strong, light carbon fiber (plastic and carbon) is great for racket frames.

Planet under pressure

Materials scientists have to be careful. Taking elements such as rare-earth metals from beneath the ground or ocean can damage the environment. Other materials that end up back in the soil, air, or water can harm wildlife.

Solving problems

New materials can solve problems. Plastic is cheap, light, and strong. However, it takes a very long time to break down, so plastic waste builds up. Scientists are trying to design types of plastic that are easier to reuse and recycle.